

Dhruv Mandal

AI & DATA SCIENCE ENGINEER

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EDUCATION

B.Tech in Artificial Intelligence & Data Science

2024 – Present

Ajeenkya DY Patil School of Engineering, Pune

- Coursework: Deep Learning, Statistical Machine Learning, Natural Language Processing, Data Engineering.
- Active contributor to college AI research initiatives and competitive hackathons.

EXPERIENCE

Machine Learning Developer & Automation Engineer

Jan 2026 – June 2026

O3GenAI IT Solutions

- Architected and deployed an end-to-end NLP classification pipeline that automated triage and tagging for 2,000+ monthly customer tickets.
- Developed high-throughput lead intelligence and automation workflows leveraging GPT-4 API and Playwright.
- Optimized prompt engineering and inference latency, cutting token consumption by 35%.

KEY PROJECTS

Lead Scoring Engine | Python, XGBoost, FastAPI, Docker

- Trained gradient-boosted classification model on 18 months of CRM interaction data to rank leads by conversion probability.
- Deployed as containerized REST microservice on FastAPI, reducing sales team outreach overhead by ~30%.

Brand Sentiment Monitor | HuggingFace, DistilBERT, Next.js, WebSockets

- Built real-time streaming analytics dashboard classifying live social media streams at 500+ items/min.
- Fine-tuned lightweight DistilBERT model delivering sub-50ms inference time with 92% sentiment classification accuracy.

Clinical Medical Document Classifier | BioBERT, PyTorch, Scikit-learn

- Fine-tuned BioBERT on 12,000 clinical medical records to categorize documents into 14 ICD-10 diagnostic categories.
- Achieved 91% macro F1 score on validation set, outperforming traditional TF-IDF baseline by 18 points.

End-to-End Churn Prediction Pipeline | MLflow, Airflow, SHAP, Python

- Constructed complete ML lifecycle pipeline engineering 40+ customer behavioral features with scheduled Airflow DAGs.
- Integrated SHAP explainability values into executive dashboards, providing actionable retention triggers.

TECHNICAL SKILLS

Languages: Python, C++, SQL, JavaScript / TypeScript

AI / ML Frameworks: PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, XGBoost

Specializations: NLP Pipelines, RAG Systems, LLM Fine-tuning, Predictive Modeling, SHAP

Tools & Infrastructure: FastAPI, Docker, Apache Airflow, MLflow, Git, Jupyter, Linux